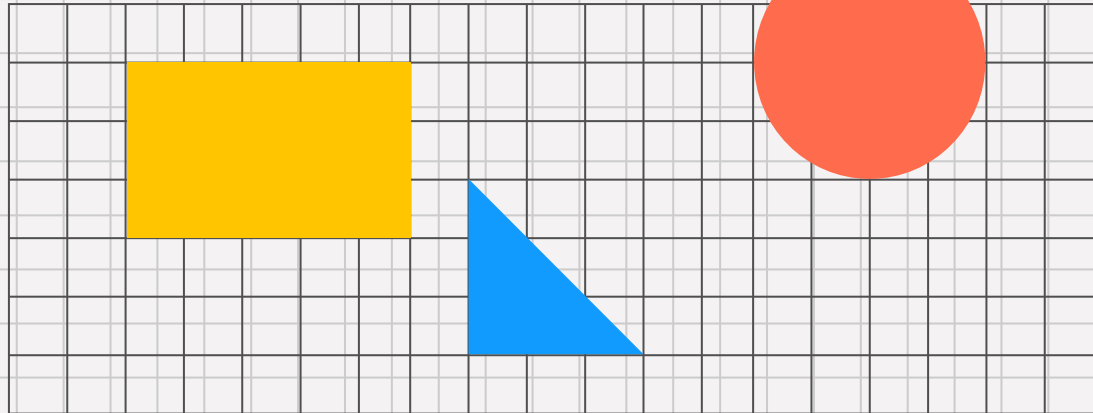


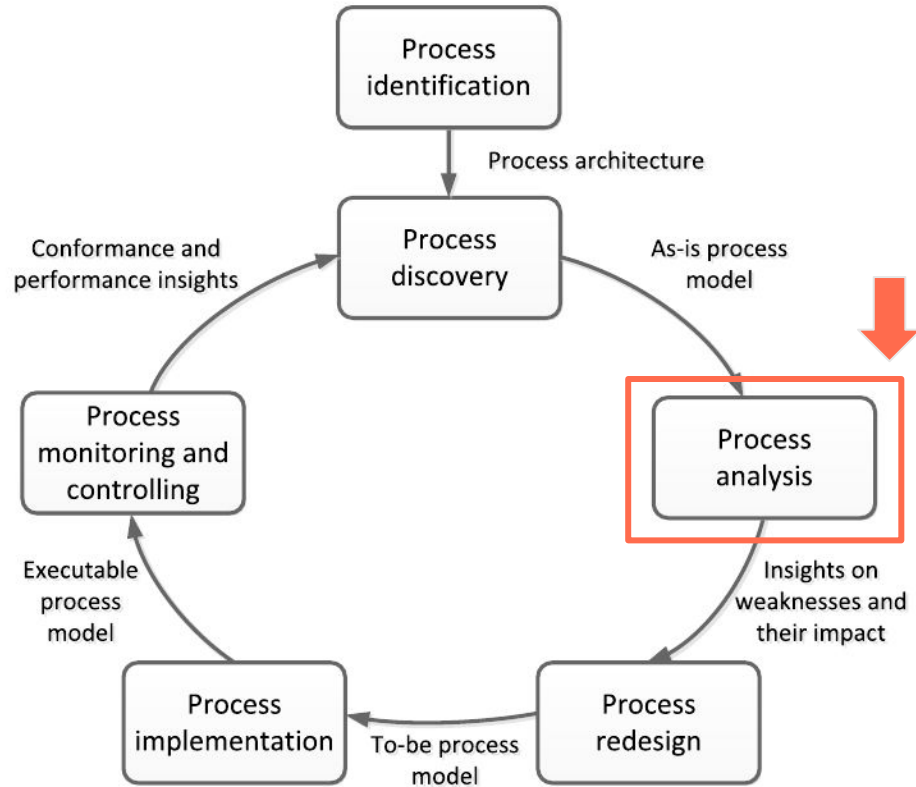
Spring 2026

IS 365

Lecture 7a:
Process Analysis (Qualitative)



Focus This Week



Qualitative Analysis

Value-Added Analysis

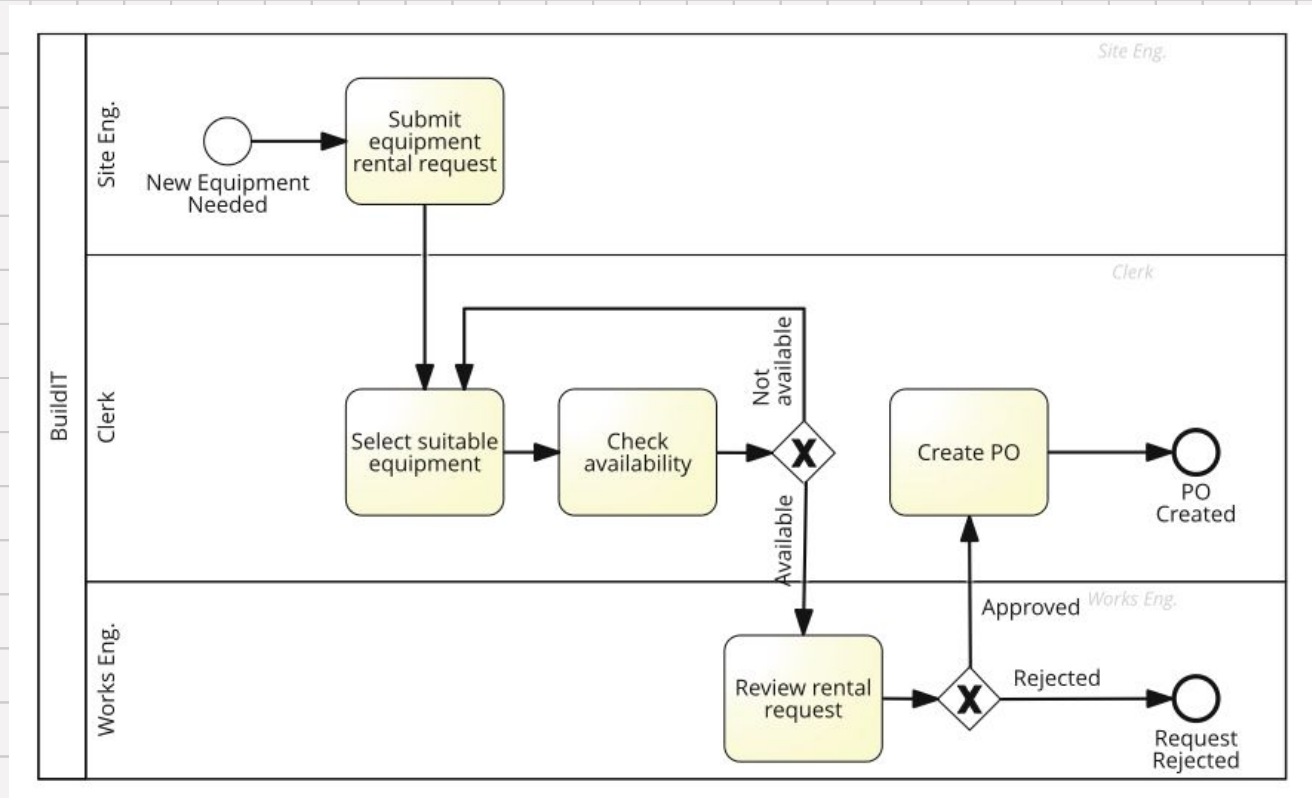
Purpose

- Identify unnecessary or redundant steps in processes
- Separate tasks into Value Adding (VA), Business Value Adding (BVA), Non-Value Adding (NVA)

3 Categories of Process Steps

- **VA:** Directly benefits customers (creates perceived value)
- **BVA:** Necessary for business or compliance reasons
- **NVA:** Pure waste — candidate for elimination

Example: Equipment Rental Process



Classification of Steps

Step	Performer	Classification
Fill request	Site engineer	VA
Send request to clerk	Site engineer	NVA
Open and read request	Clerk	NVA
Select suitable equipment	Clerk	VA
Check equipment availability	Clerk	VA
Record recommended equipment & supplier	Clerk	BVA
Forward request to works engineer	Clerk	NVA
Open and examine request	Works engineer	BVA
Communicate issues	Works engineer	BVA
Forward request back to clerk	Works engineer	NVA
Create PO	Clerk	BVA
Send PO to supplier	Clerk	VA

Reducing NVA

- Automation (e.g., digital forms)
- Policy simplification
- Process redesign

Waste Analysis

Introduction to Waste (Muda)

- **Waste:** activities consuming resources without creating value
- **Muda** is the Japanese word for “waste”
- Categories of waste come from **Japanese lean thinking**

Types of Waste

- **Move:** unnecessary transportation, hand-offs, motion
- **Hold:** waiting, inventory buildup
- **Overdo:** defects, rework, overproduction, overprocessing

Lean manufacturing 8 wastes with examples

Over production	Waiting	Motion	Transportation
			
Batch too big	Wait for materials	Poor ergonomics	Move products
Over processing	Inventory	Defects	Unused talents
			
Too many steps	Work in Progress	Defective product	Inadequate training

Examples of Process Waste

- Excessive approvals → overprocessing
- Bottlenecks → waiting waste
- Rework due to errors → defect waste

Goal of Waste Analysis

- Eliminate or minimize waste categories
- Align process with customer value

Stakeholder & Issue Analysis

Stakeholder Perspectives

- **Stakeholders:** customers, employees, managers, regulators
- Each group provides different process pain points

Issue Register

- **Structured list of issues with fields:**
description, priority, data assumptions
- Helps systematically track and prioritize problems

Example:

Issue 1

Issue 1: Equipment kept longer than needed

Priority: 1

Description: Site engineers keep the equipment longer than needed

Data and assumptions: BuildIT rents 3,000 pieces of equipment per year. In 10% of cases, site engineers keep the equipment 2 days longer than needed. On average, rented equipment costs € 100 per day

Qualitative impact: Not applicable

Quantitative impact: $0.1 \times 3,000 \times 2 \times 100 = \text{€ } 60,000$ in additional rental expenses per year

Example:

Issue 2

Issue 2: Rejected equipment

Priority: 2

Description: Site engineers sometimes reject the delivered equipment due to non-conformance to their specifications

Data and assumptions: BuildIT rents 3,000 pieces of equipment per year. Each time an equipment is rejected due to a mistake on BuildIT's side, BuildIT is billed the cost of 1 day of rental, that is € 100. 5% of them are rejected due to an internal mistake within BuildIT (as opposed to a supplier mistake)

Qualitative impact: These events disrupt the construction schedules and create frustration and internal conflicts

Quantitative impact: $3,000 \times 0.05 \times 100 = \text{€ } 15,000$ per year

Example:

Issue 3

Issue 3: Late payment fees

Priority: 3

Description: BuildIT pays late payment fees because invoices are not paid by the due date

Data and assumptions: BuildIT rents 3,000 pieces of equipment per year. Each equipment is rented on average for 4 days at a rate of € 100 per day. Each rental leads to one invoice. About 10% of invoices are paid late. On average, the penalty for late payment is 2% of the amount of the invoice

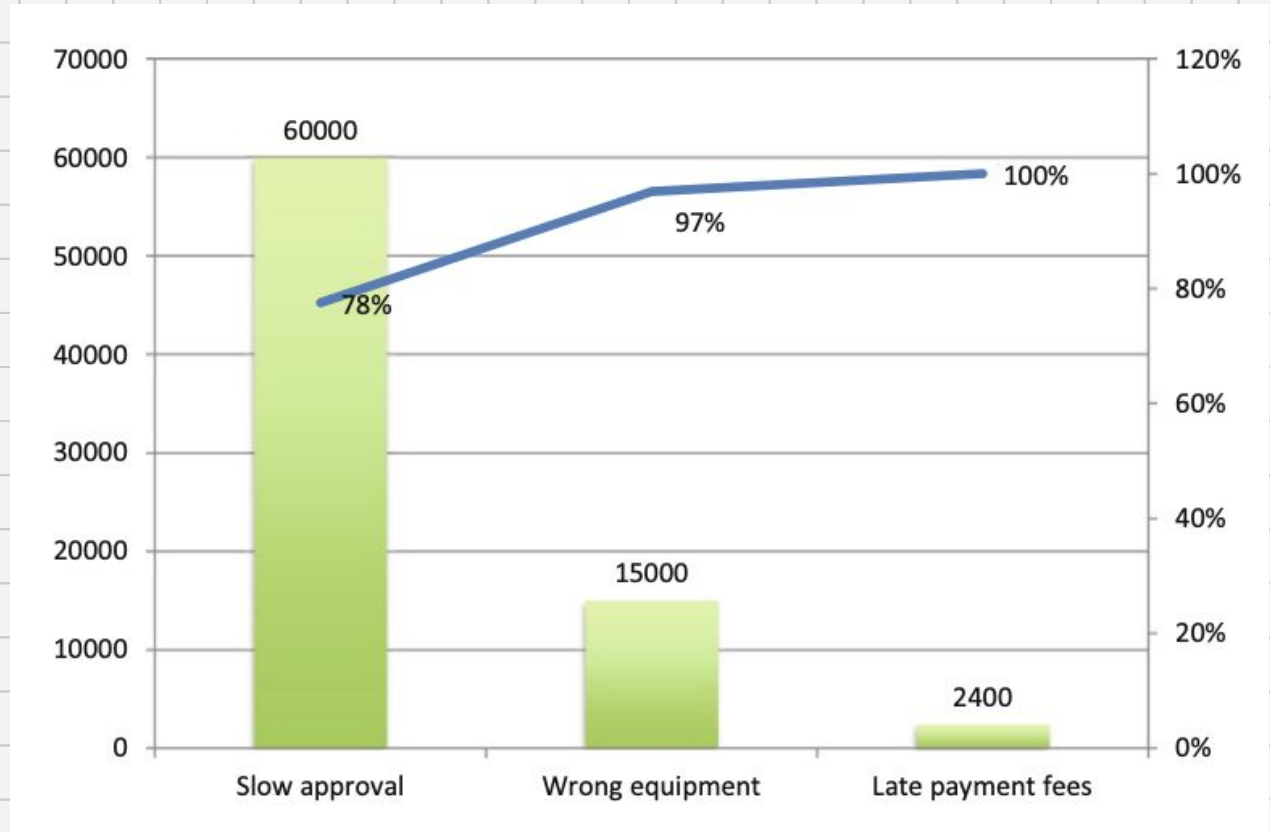
Qualitative impact: Suppliers are annoyed and later unwilling to negotiate more favorable terms for equipment rental

Quantitative impact: $0.1 \times 3,000 \times 4 \times 100 \times 0.02 = \text{€ } 2,400$ per year

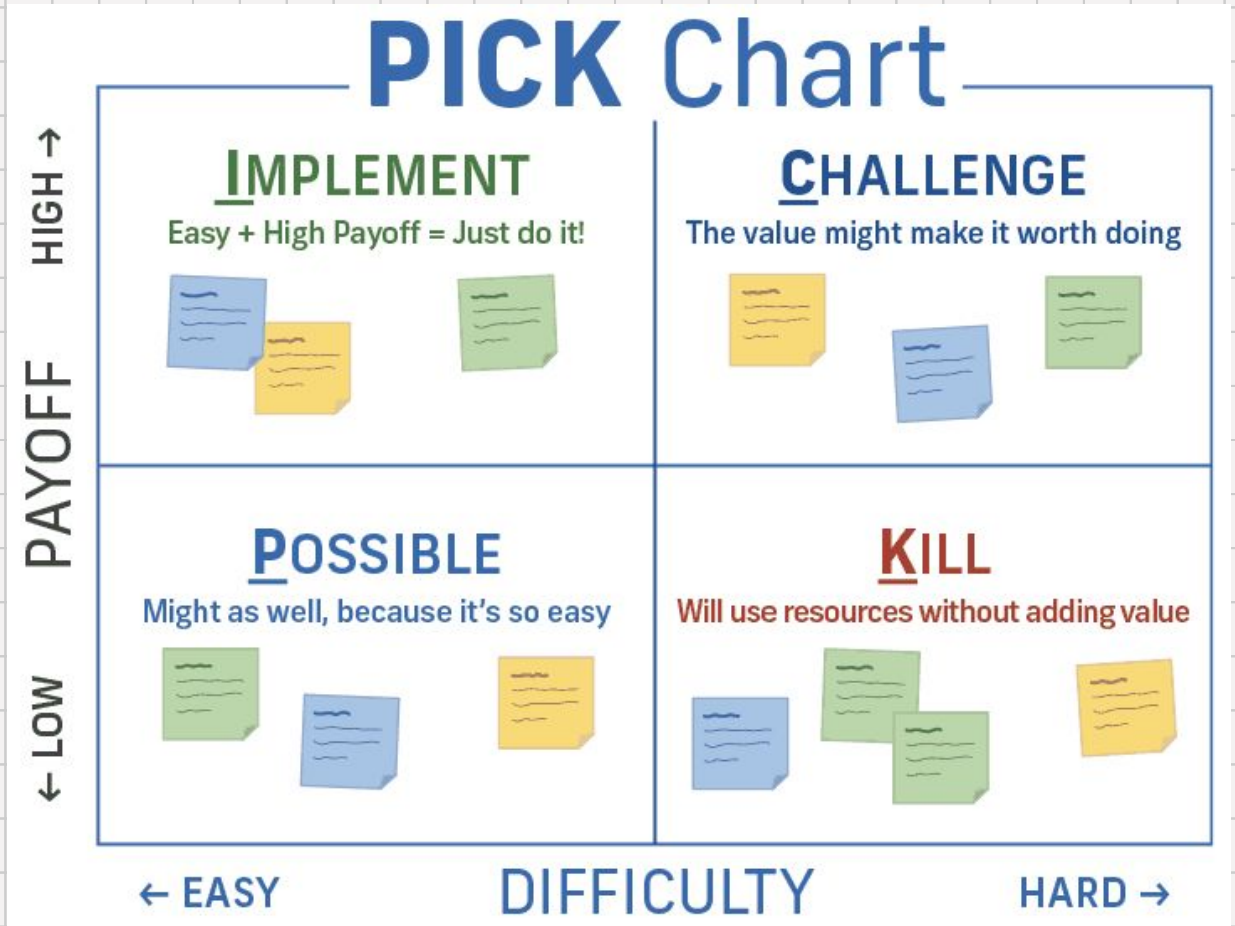
Prioritization Tools

- **Pareto principle:** focus on top 20% issues with 80% impact
- **PICK chart:** payoff vs difficulty analysis

Pareto Chart



PICK Chart



Root Cause Analysis

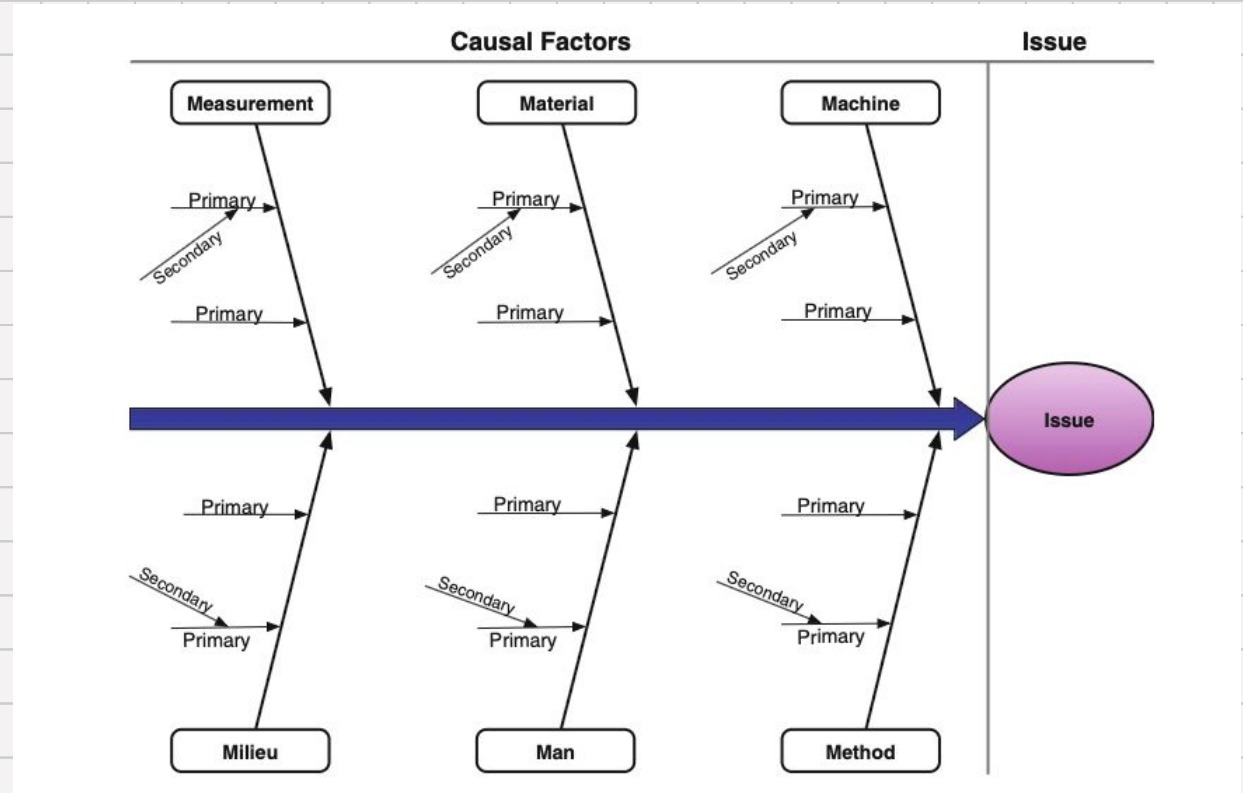
Why Root Cause?

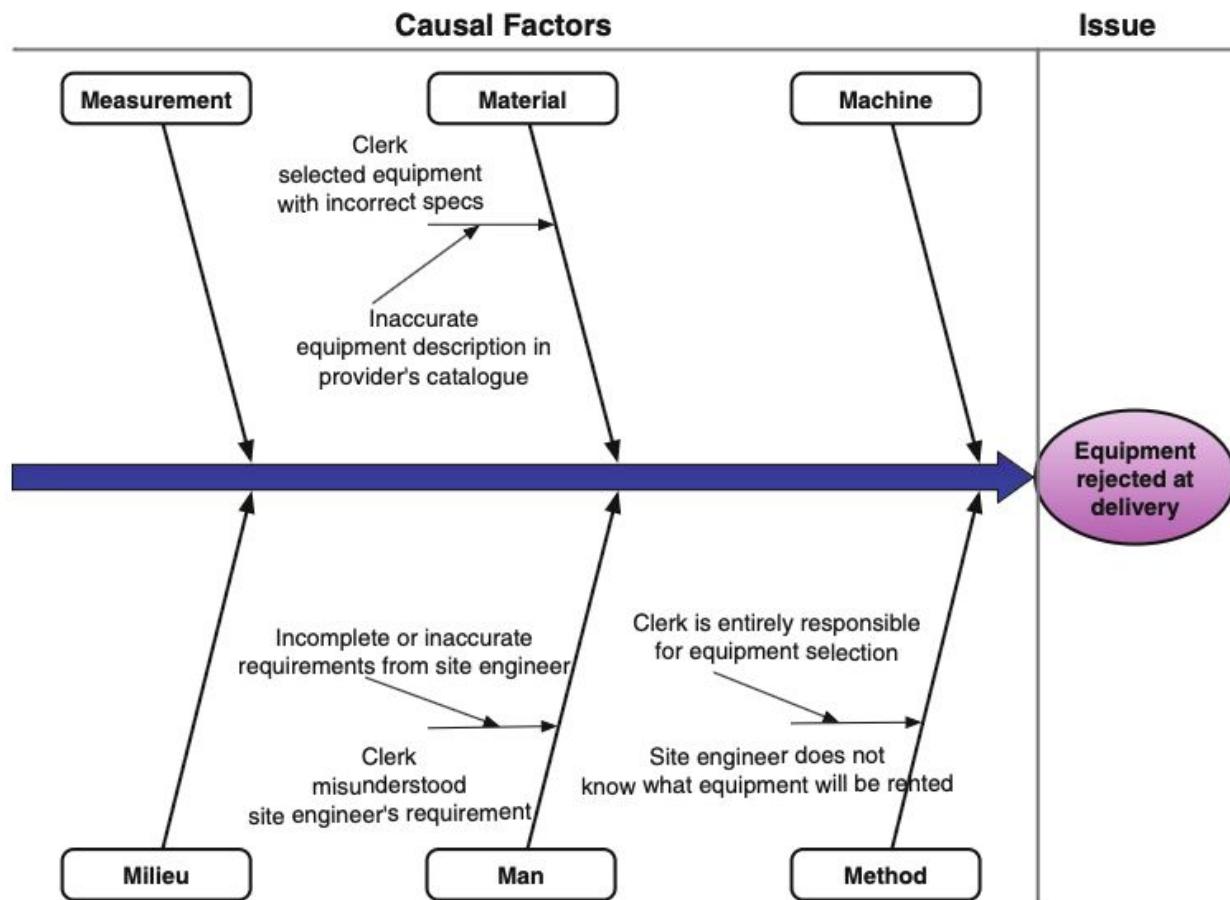
- Avoids treating symptoms; focuses on underlying problems
- Helps in designing sustainable process improvements

Fishbone Diagram

- Cause-effect diagram (Ishikawa) groups causes into categories
- **6Ms:** Machine, Method, Material, Man, Measurement, Milieu

Ishikawa Chart

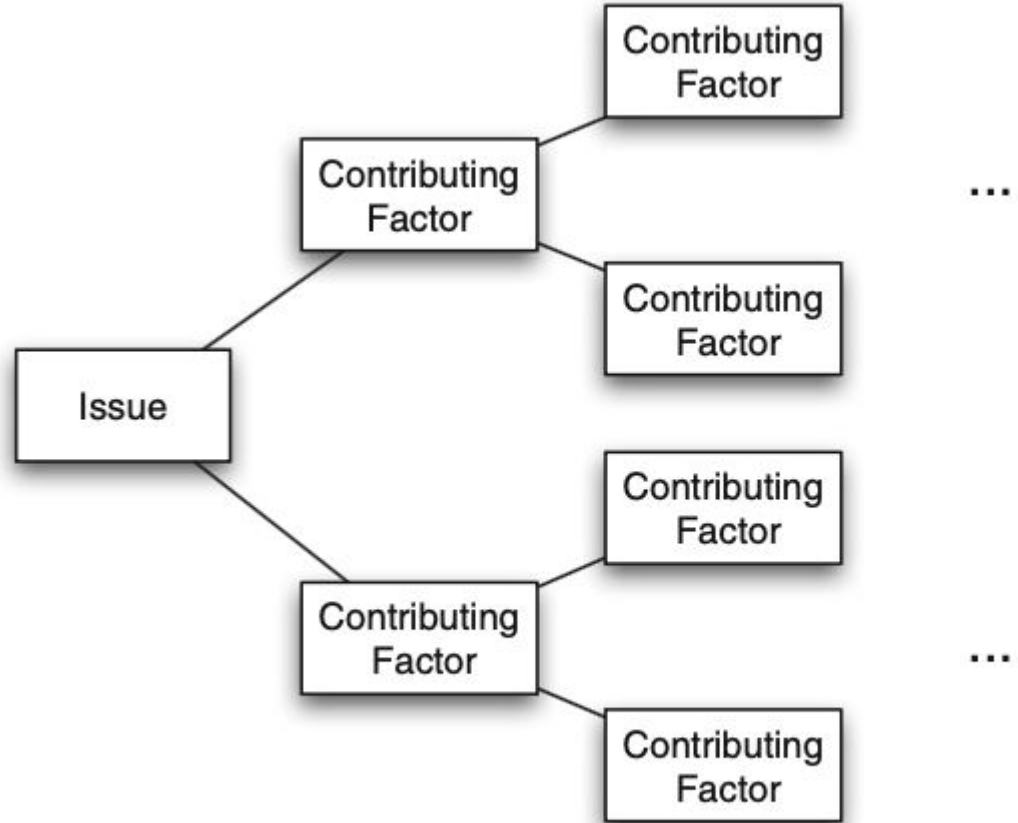




Why-Why Analysis

- Recursive questioning technique (ask 'why?' 5 times)
- Drills down to the fundamental reason for issues

Why-Why Analysis Template



Issue 1 Site engineers sometimes reject delivered equipment, why?

- wrong equipment is delivered, why?
 - miscommunication between site engineer and clerk, why?
 - site engineer provides an incomplete or inaccurate description of what they want.
 - site engineer does not always see the supplier catalogs when making a request and does not communicate with the supplier, why?
 - site engineer generally does not have Internet connectivity.
 - site engineer does not check the choice of equipment made by the clerk.
 - equipment descriptions in supplier's catalog not accurate.

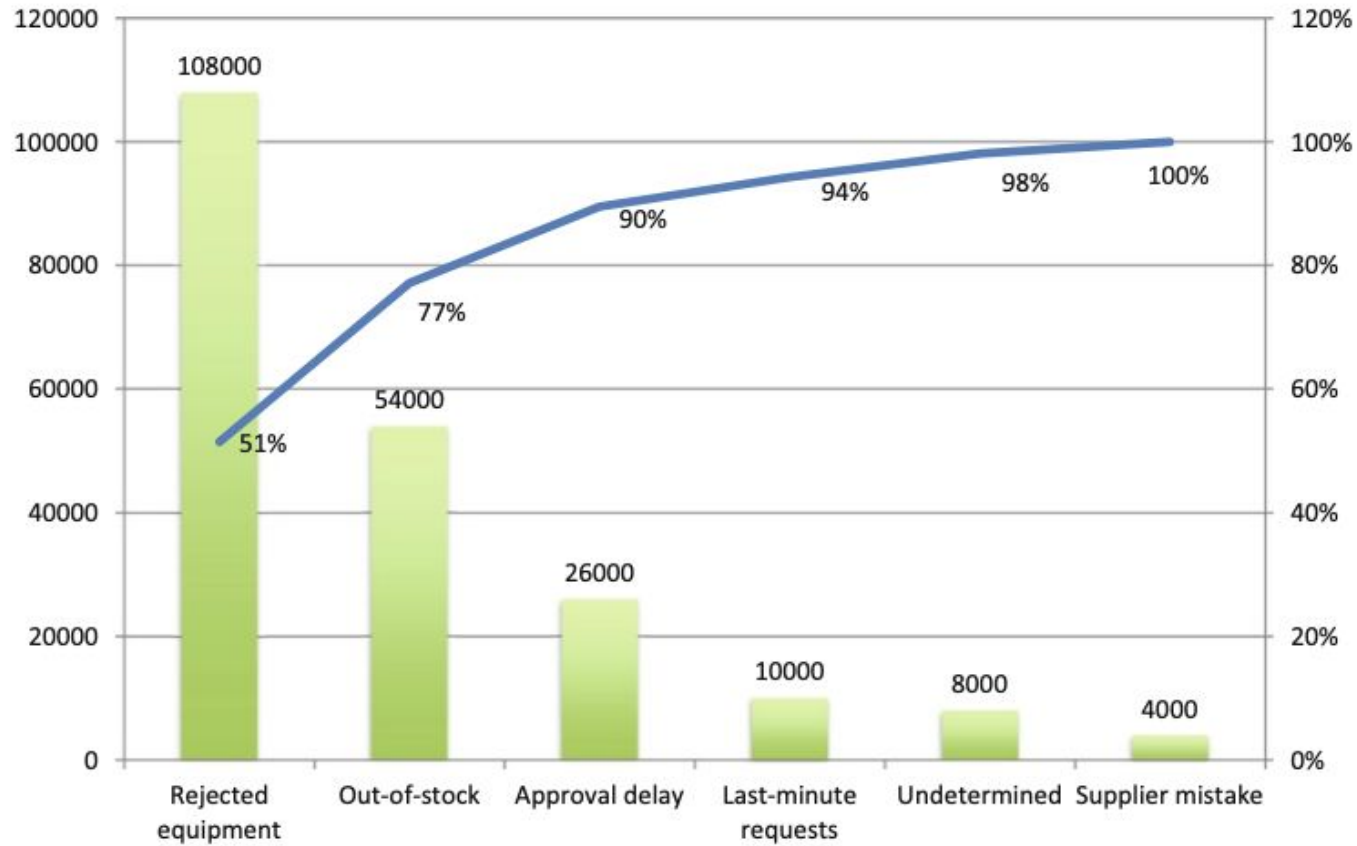


Fig. 6.8 Pareto chart of causal factors of issue "Equipment not available when needed"